# Embeddings and Semantic Similarity

*Representing meaning for vector search*

## Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

## 1. What Is an Embedding?

An embedding is a numerical vector produced from text, images, or another input. Text embeddings are designed so that useful semantic relationships can be represented geometrically. Passages about related ideas may have vectors that are close under a chosen similarity or distance measure. An embedding is not a human-readable definition and its individual dimensions usually do not correspond to simple concepts.

## 2. Creating Document and Query Vectors

A retrieval pipeline typically splits documents into chunks, creates an embedding for each chunk, and stores the vector with the chunk text and metadata. At query time, the user's question is embedded using a compatible model. The system compares the query vector with stored vectors and returns the most similar chunks. Query and document vectors should be generated with compatible model versions and expected input formats.

## 3. Cosine Similarity

Cosine similarity compares the angle between two vectors: $\cos(\theta) = (A \cdot B) / (||A|| \, ||B||)$. A value closer to 1 indicates that vectors point in similar directions; values near 0 indicate little angular alignment; negative values indicate opposing directions. The interpretation depends on the model and dataset. Some vector databases use cosine distance, dot product, or Euclidean distance, and scores should not be compared across systems without understanding the metric.

## 4. Semantic Similarity Is Not Truth

A passage can be semantically similar to a question yet fail to answer it. A question about "the risks of fine-tuning" may retrieve a passage that only defines fine-tuning. Conversely, an exact keyword match can be important even if its overall semantic score is modest. Retrieval systems may combine semantic search with lexical search, metadata filters, reranking, or domain-specific rules to improve relevance.

## 5. Embedding Model Choices

Consider domain coverage, language support, vector dimension, maximum input length, latency, cost, and licensing. A larger vector dimension is not automatically better. Model updates can change vector spaces, so vectors generated by incompatible versions should not be mixed casually. If an embedding model changes, re-embedding the corpus may be necessary. Store model identity and ingestion configuration with the dataset when reproducibility matters.

## 6. Practical Debugging

If search results look poor, inspect the exact extracted chunk, the query text, the similarity metric, and the returned metadata. Check for empty chunks, duplicated text, truncated passages, and inconsistent preprocessing. Compare results for paraphrased questions and exact terminology. Record the top-k results for a small fixed evaluation set so changes to the embedding model or chunking strategy can be measured instead of judged only by intuition.

## Key Takeaways

Embeddings provide a geometric representation that makes semantic search possible. Similarity scores rank candidates; they do not prove that a passage is correct, complete, or sufficient to answer the question.

End of document • Use source passages as evidence and verify claims during evaluation.